

1. Research Problem and Context

Urban areas are progressively reliant on environmental forecasting for the purpose of event planning, public health protection, and the management of urban operations. Air pollution episodes, heatwaves, and adverse weather conditions can significantly affect outdoor activities, particularly large-scale events such as marathons, festivals, school programmes, and community gatherings. The toxicity of PM_{2.5} is particularly problematic due to its capacity to penetrate deep into the lungs and bloodstream, thereby contributing to the development of cardiovascular and respiratory diseases (WHO, 2021). As Kunak (2024) and Clarity (2023) have demonstrated, urban PM_{2.5} concentrations are significantly impacted by traffic emissions, meteorological conditions, and temporal patterns associated with human activity.

Despite advances in environmental monitoring, cities still lack forecasting systems that integrate multiple data modalities, such as traffic patterns, weather conditions, and city planning calendars, to produce accurate, localised PM_{2.5} predictions. In urban environments, traffic is a significant contributor to PM_{2.5} levels. Furthermore, congestion spikes associated with events, road closures or school schedules can have a substantial impact on pollution levels. However, these planning-related signals are seldom incorporated into forecasting models, despite their predictability and public availability.

Concurrently, city planners and event organisers frequently encounter difficulties in interpreting raw pollution forecasts. As the draft itself stated, "forecasting models provide numbers, not decisions." In the absence of contextual interpretation, numerical PM_{2.5} predictions are rendered ineffectual and lack the potential to provide actionable guidance. The UKHSA and WHO provide health guidelines based on thresholds; however, in order to apply these to specific events, participant sensitivities and local conditions, it is necessary to employ a reasoning process that is not provided by traditional forecasting systems.

The present research aims to address these gaps by developing:

1. A multi-modal PM_{2.5} forecasting model using traffic, weather and average energy consumption. (I haven't found historical bath planning calendar, but I want to include it for evaluation of the system with real planned events)
2. A separate LLM-based MCP advisor that interprets the forecasts against UK/EU standards to generate structured risk assessments for planned events.

"Model Context Protocol (MCP), a new standard for connecting AI assistants to the systems where data lives, including content repositories, business tools, and development environments. Its aim is to help frontier models produce better, more relevant responses." (Anthropic 2024)

Together, these components aim to bridge the gap between environmental forecasting and operational decision-making in urban contexts.